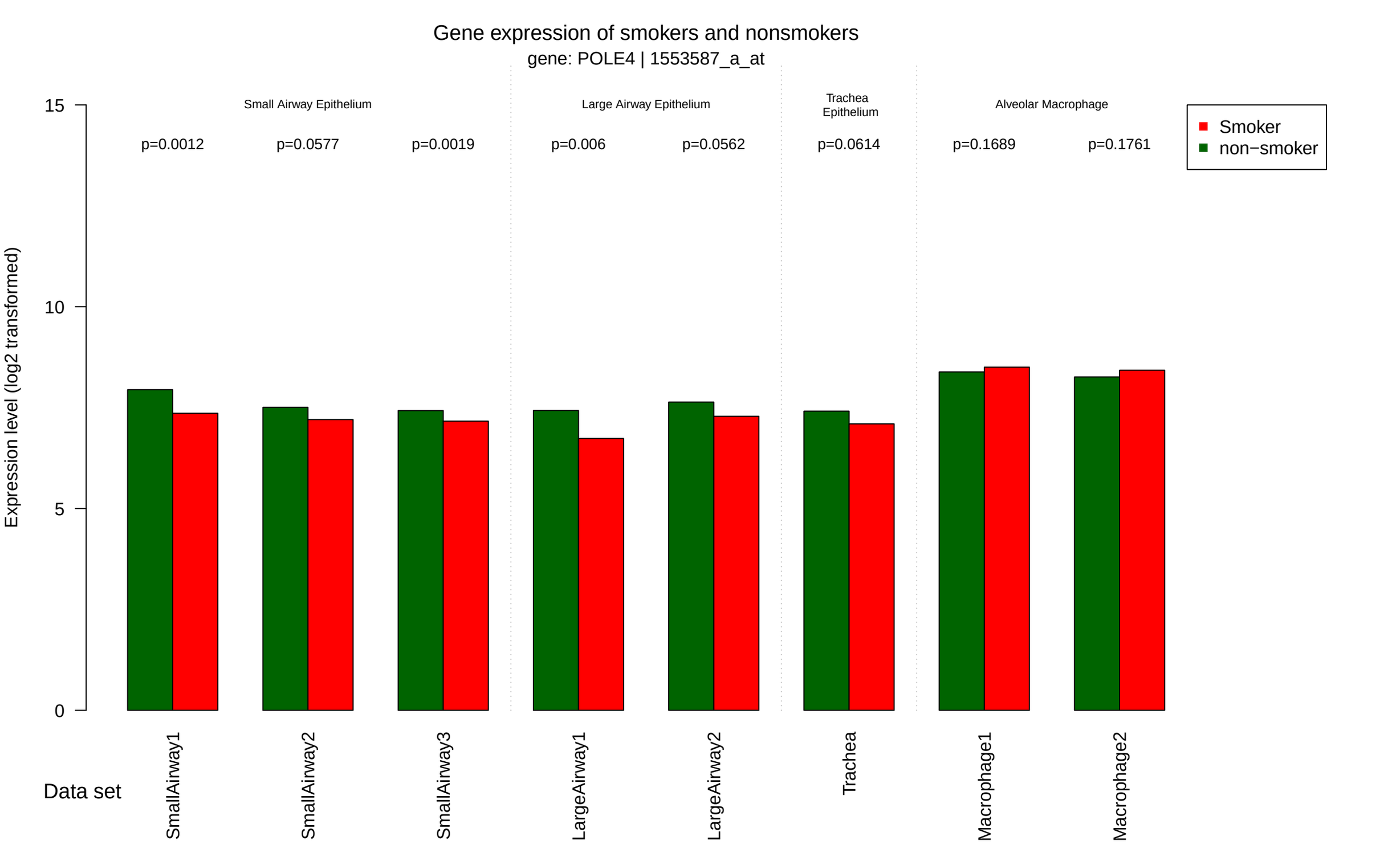
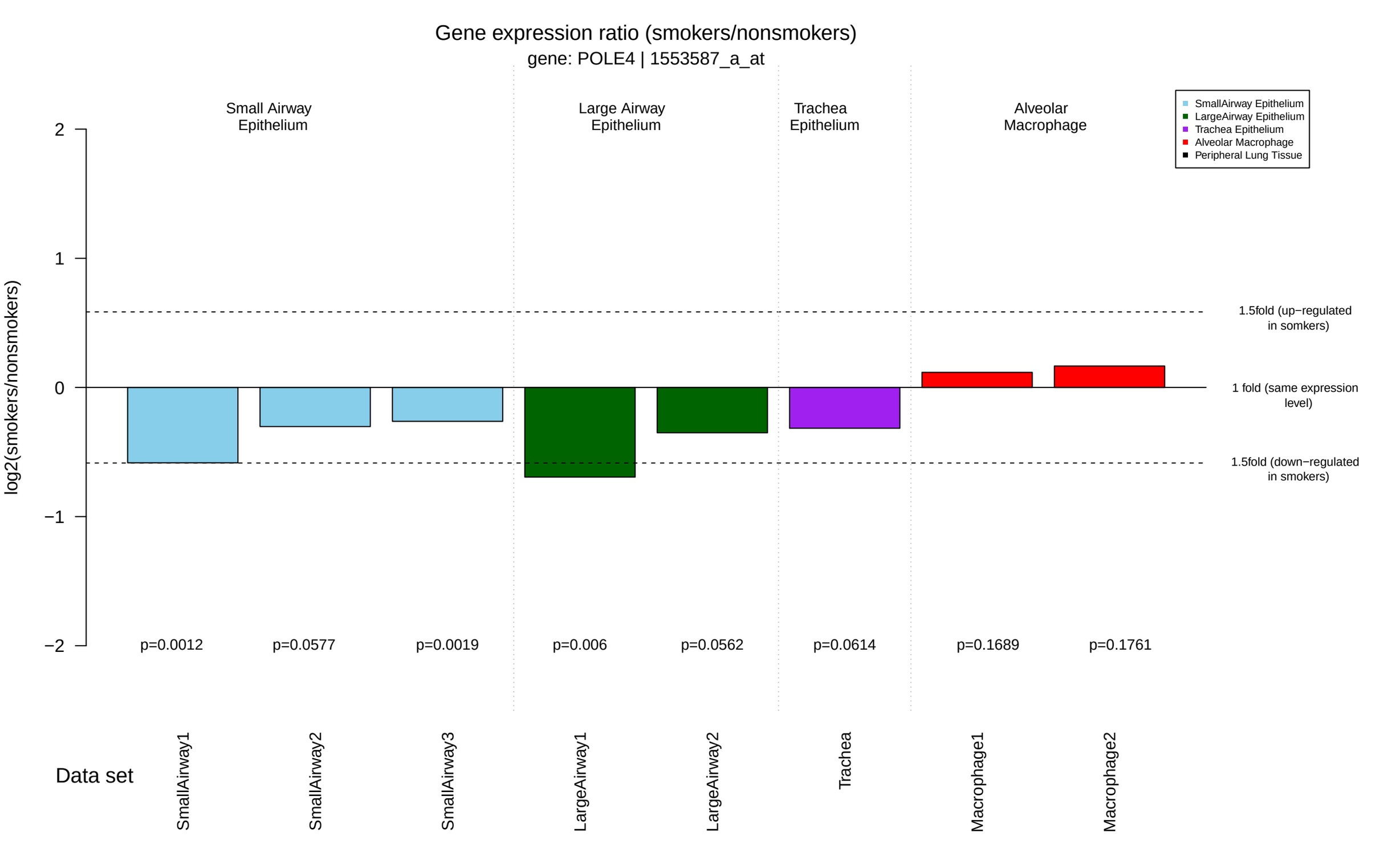
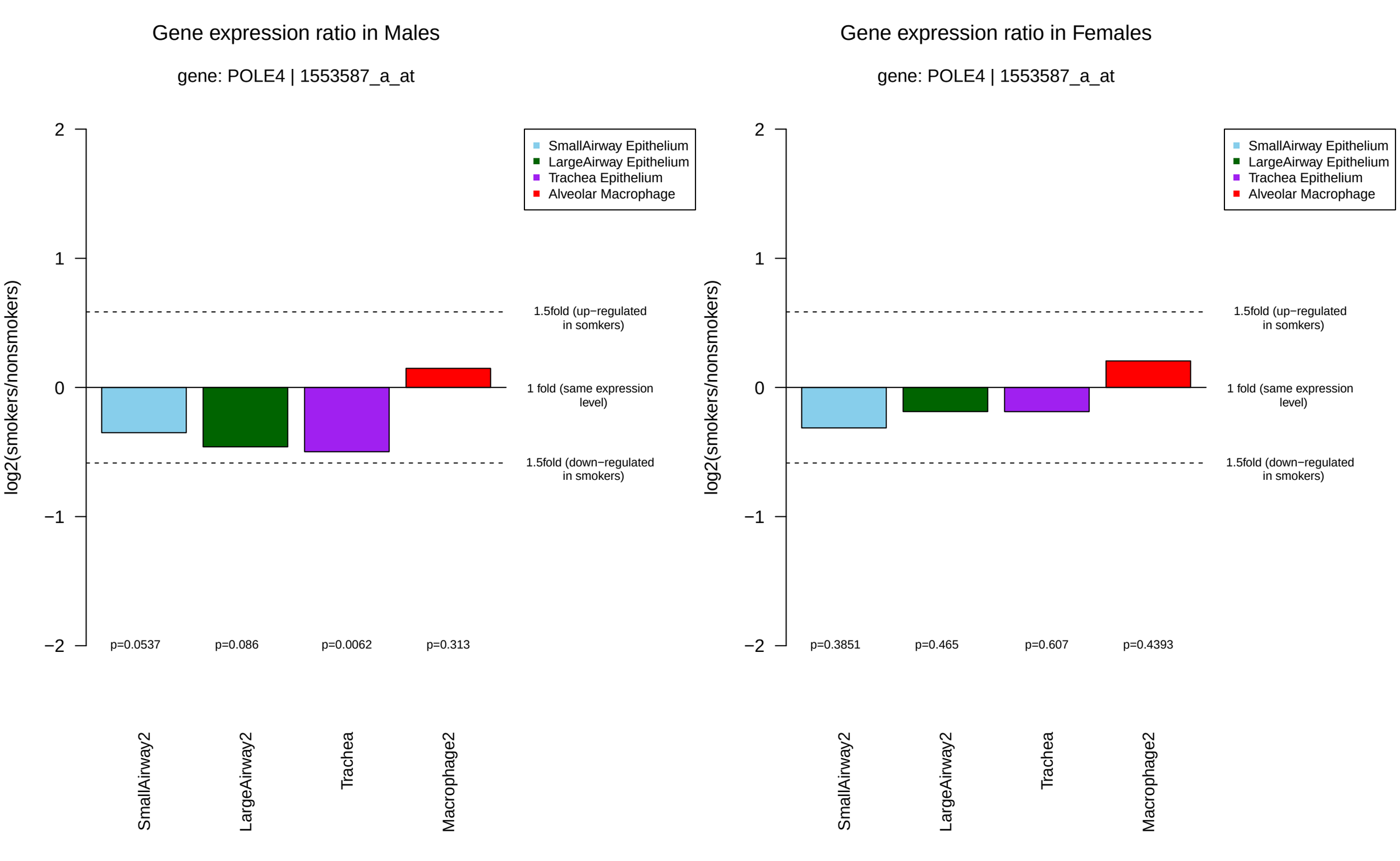
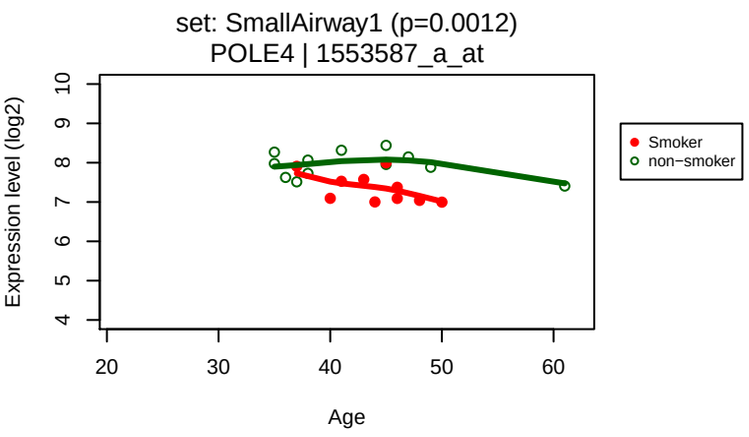
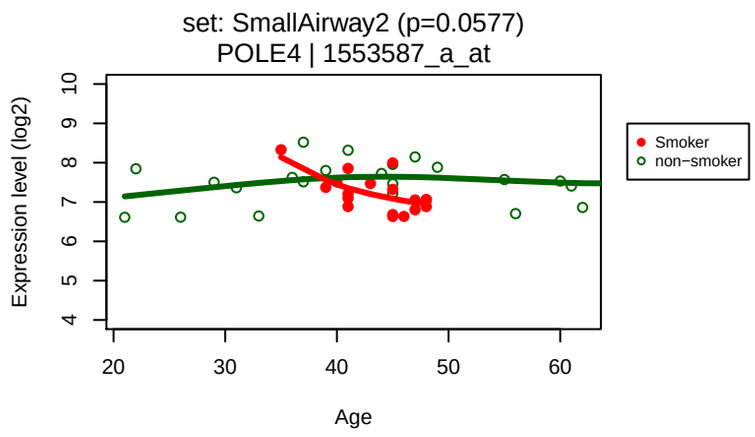
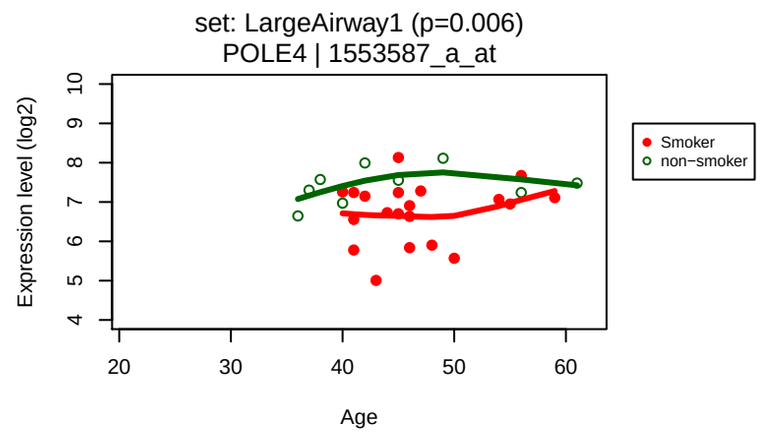

Gene expression of smokers and nonsmokers
gene: POLE4 | 1553587_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0012
p=0.0577
p=0.0019
p=0.006
p=0.0562
p=0.0614
p=0.1689
p=0.1761
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

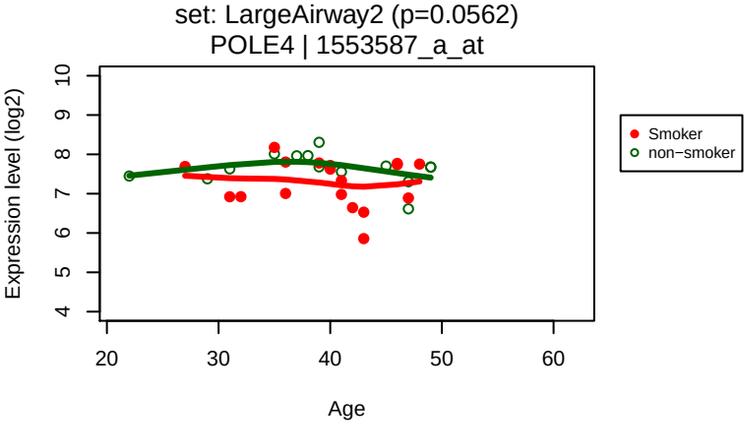

Gene expression ratio (smokers/nonsmokers)
gene: POLE4 | 1553587_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0012
p=0.0577
p=0.0019
p=0.006
p=0.0562
p=0.0614
p=0.1689
p=0.1761
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

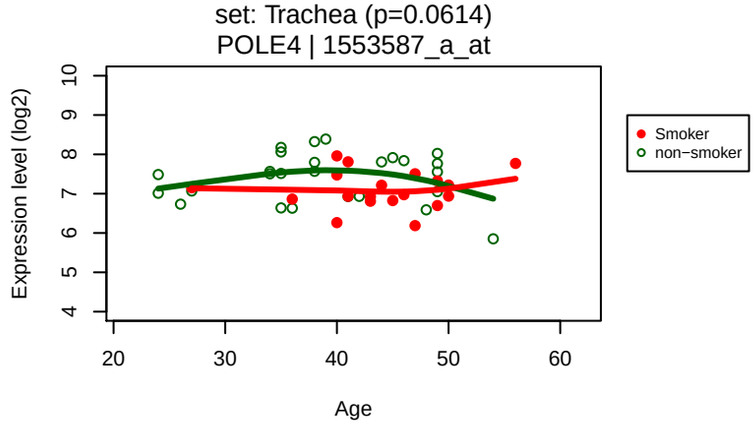

Gene expression ratio in Males
gene: POLE4 | 1553587_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0537
p=0.086
p=0.0062
p=0.313
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: POLE4 | 1553587_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3851
p=0.465
p=0.607
p=0.4393
SmallAirway2
LargeAirway2
Trachea
Macrophage2

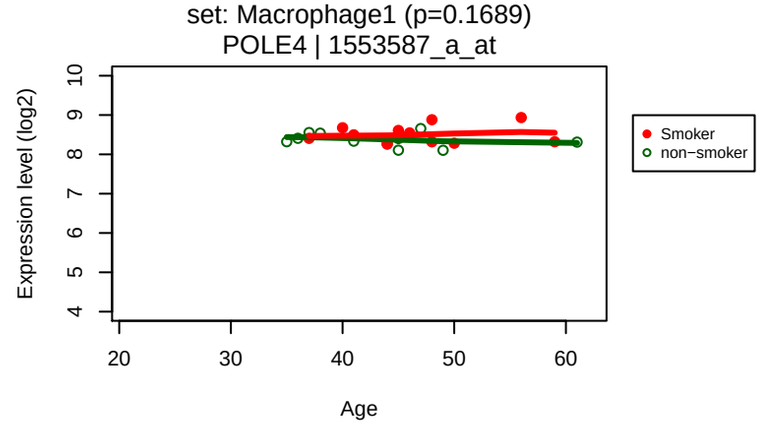

set: SmallAirway1 (p=0.0012)
POLE4 | 1553587_a_at
Expression level (log2)
Age
Smoker
non−smoker

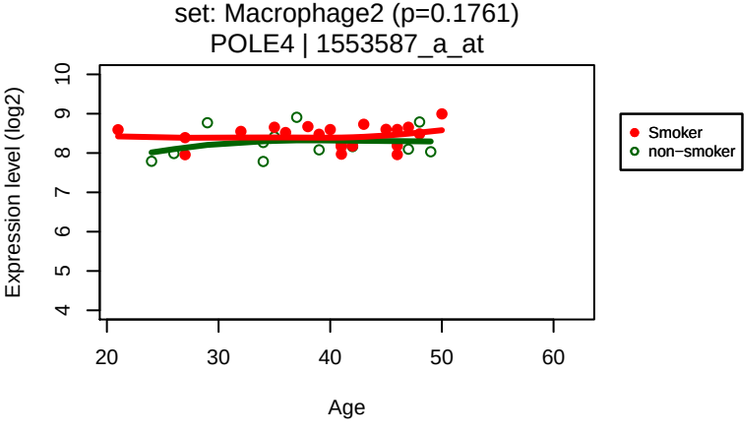

set: SmallAirway2 (p=0.0577)
POLE4 | 1553587_a_at
Expression level (log2)
Age
Smoker
non−smoker

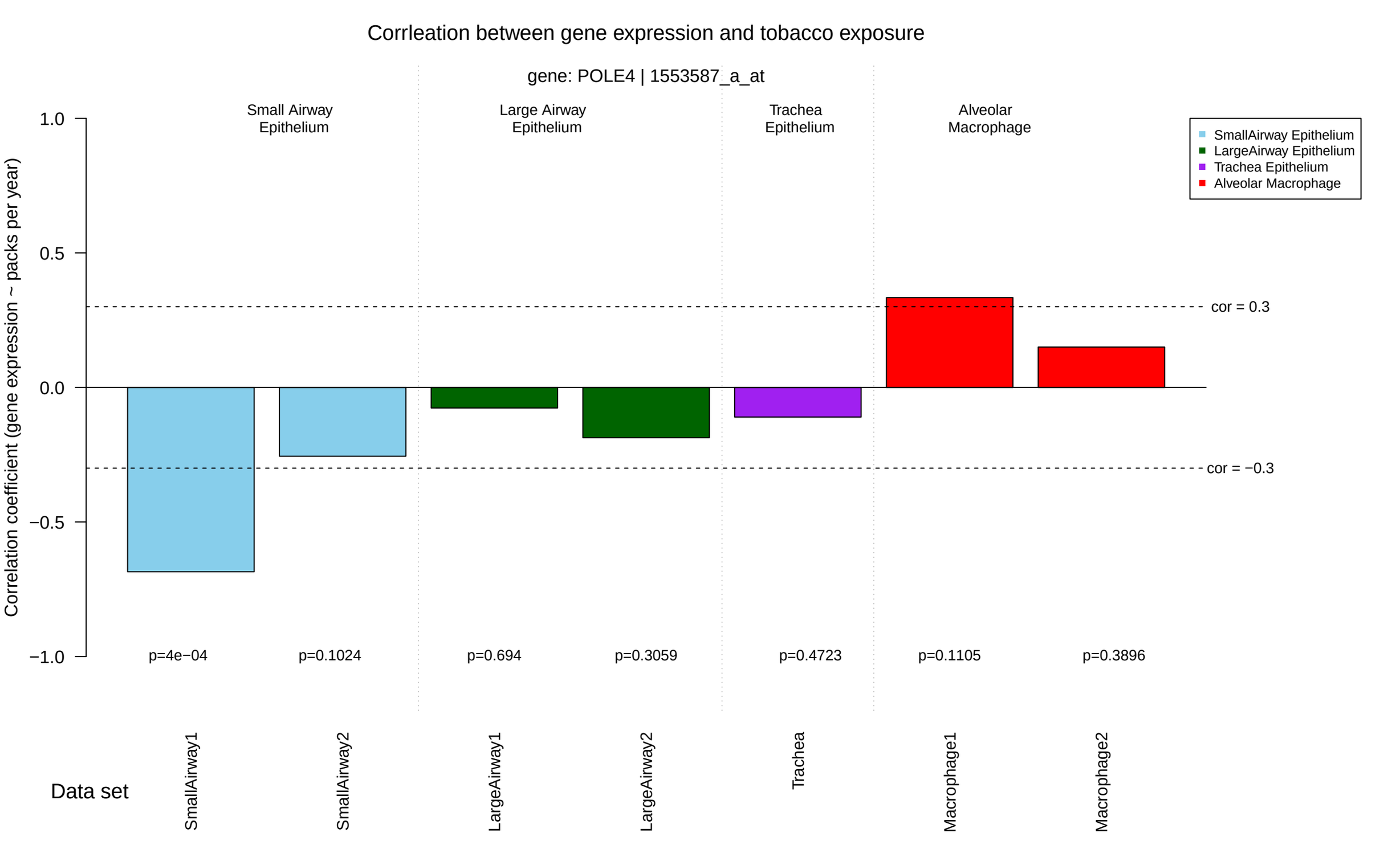

set: LargeAirway1 (p=0.006)
POLE4 | 1553587_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0562)
POLE4 | 1553587_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0614)
POLE4 | 1553587_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1689)
POLE4 | 1553587_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1761)
POLE4 | 1553587_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: POLE4 | 1553587_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=4e−04
p=0.1024
p=0.694
p=0.3059
p=0.4723
p=0.1105
p=0.3896
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2